

Object Detection Pipeline Documentation

A 7-step HPC-backed pipeline for few-shot object detection using class supports, SAM3-assisted annotation, and Tapis job execution.

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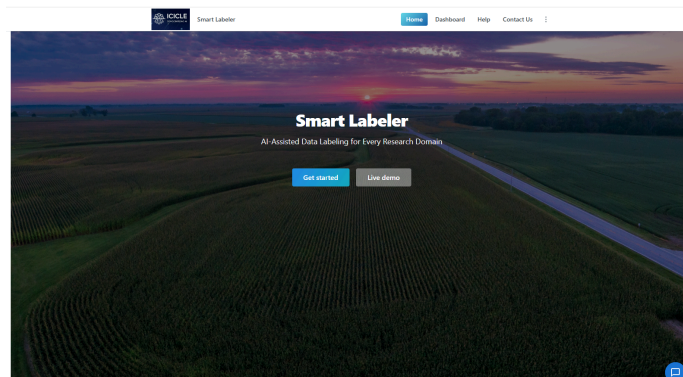
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Quick Start

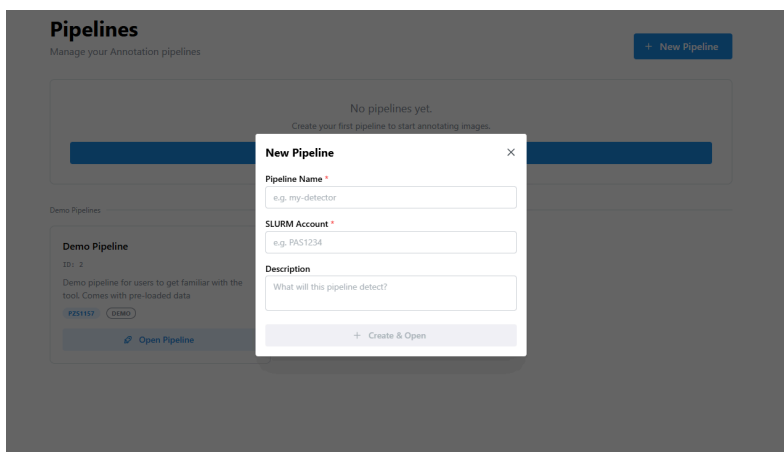
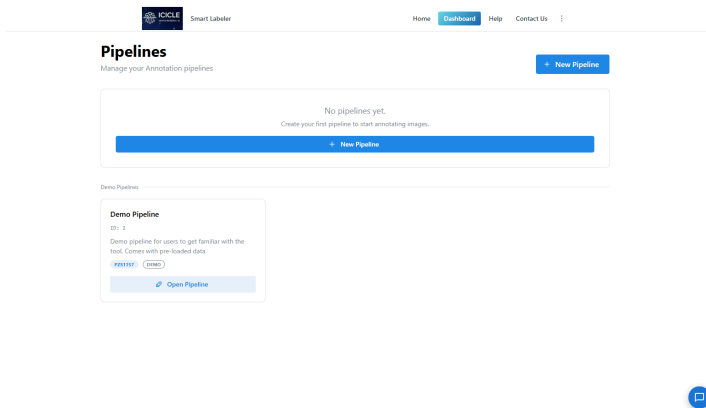
1. **Access the Application:** Navigate to the home page and select "Get Started".
 2. **Create Pipeline:** Enter a pipeline name, select jobs, and specify a Slurm account.
 3. **Follow the 7 Steps:** Use the navigation bar to progress through annotation, model training, and detection.
 4. **Monitor Progress:** Check job status via the status bar and Tapis logs.
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Home Page & Dashboard

The Home Page serves as the entry point where users can access the Dashboard or view demo video.

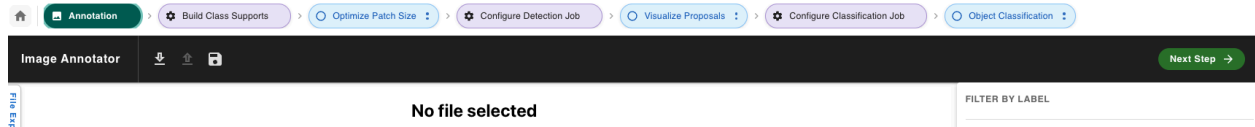


The Dashboard allows users to manage their annotation pipelines. You can create a new pipeline by providing a name, Slurm account, and optional description.

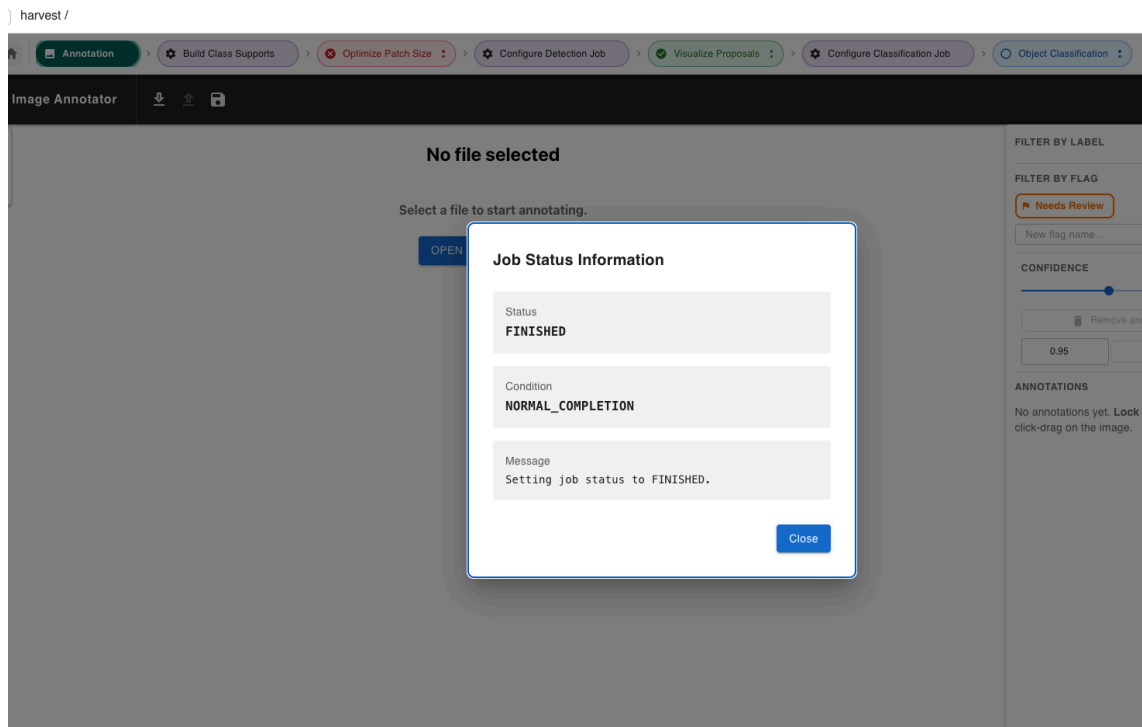
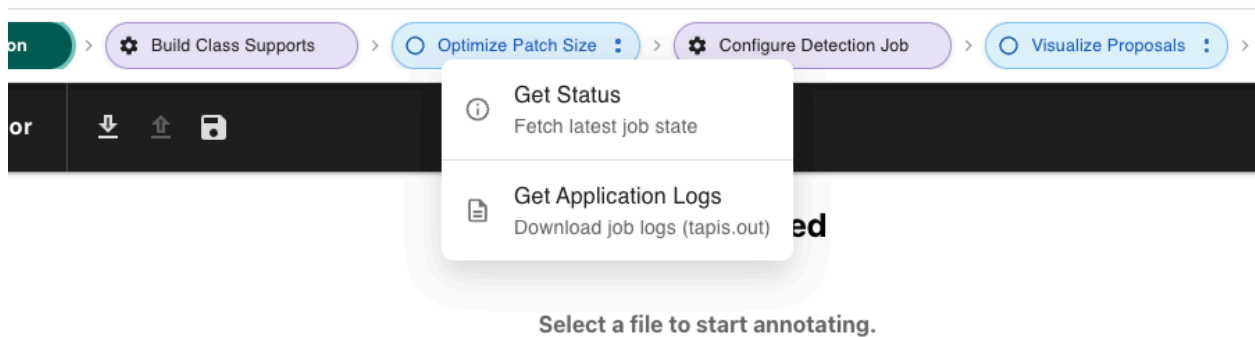


Pipeline Navigation Bar

A persistent **StepperStatusBar** appears at the top of every pipeline step, showing progress through all seven stages. Each step chip is clickable for direct navigation.

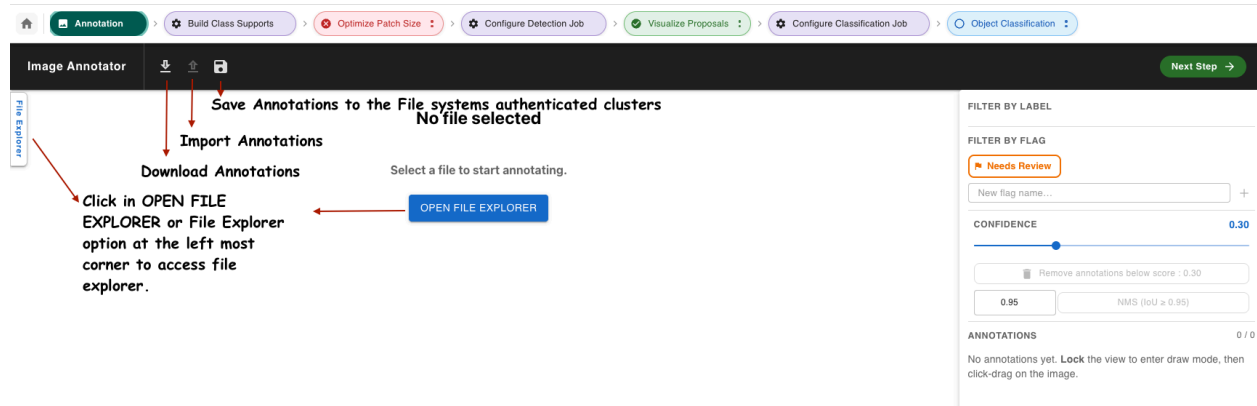


Right-clicking a step chip exposes actions such as "Get Status" (polling Tapis job status) and "Get Tapis Logs".

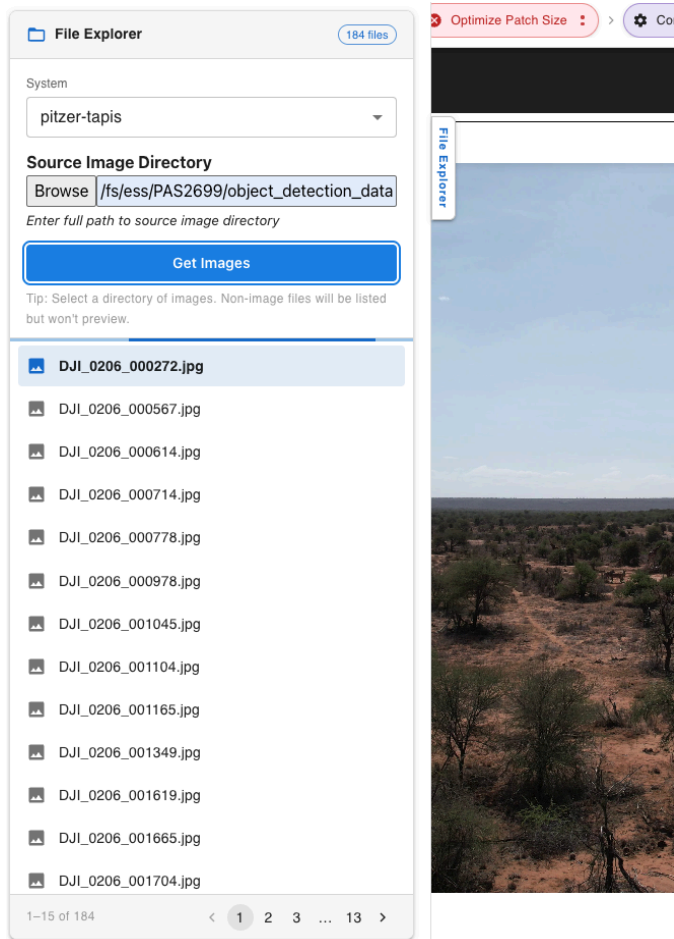


Step 1 — Image Annotator

The Image Annotator is the primary tool for creating and managing ground-truth bounding boxes. It features a File Explorer to browse remote filesystems (e.g., Pitzer, Expanse) and load image thumbnails.



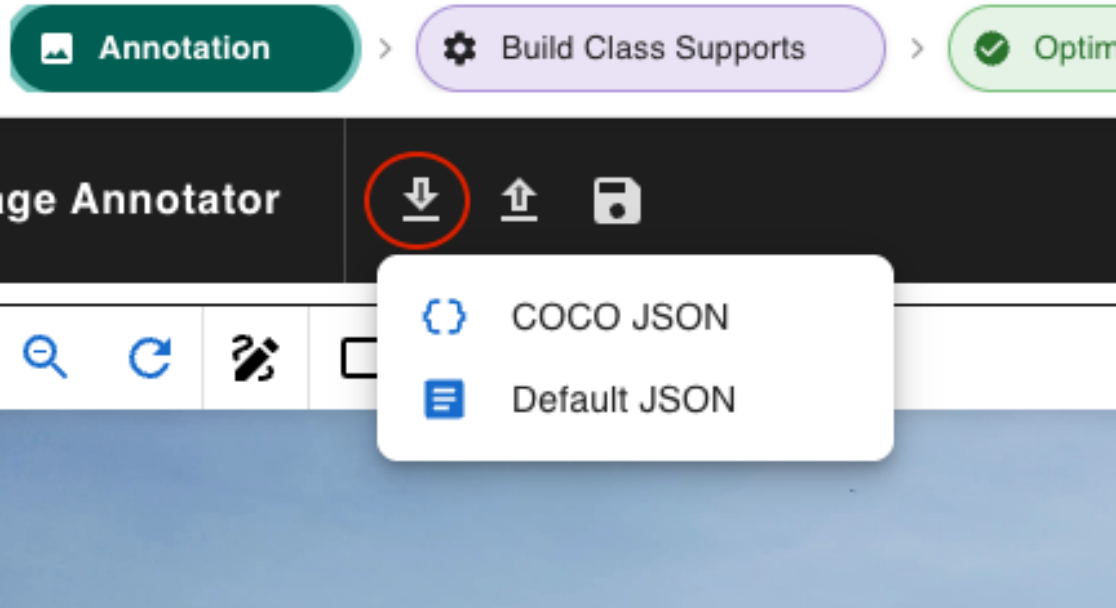
Here you can access the images you want to annotate



Comprehensive list of the tools available in the annotator



Download, Import, and Save Annotations



Import Annotations

Ways to import annotations

pitzer-tapis

Cluster

Upload File

BROWSE FILE

If annotation file is obtained from your computer user this option

Or Enter File Path

/path/to/annotations.json

☐ Default JSON

We have two format options (switch to the required option):
1. Coco Json
2. Tool Specific json

Format Preview

```
{
  "image_path": "/path/to/image.jpg",
  "class": "donkey",
  "bounding_box": [x1, y1, x2, y2]
}
```

(Just a format review)

CANCEL

UPLOAD

Save Annotations

pitzer-tapis

File Path

/path/to/save/annotations.json

☒ COCO JSON

Format Preview

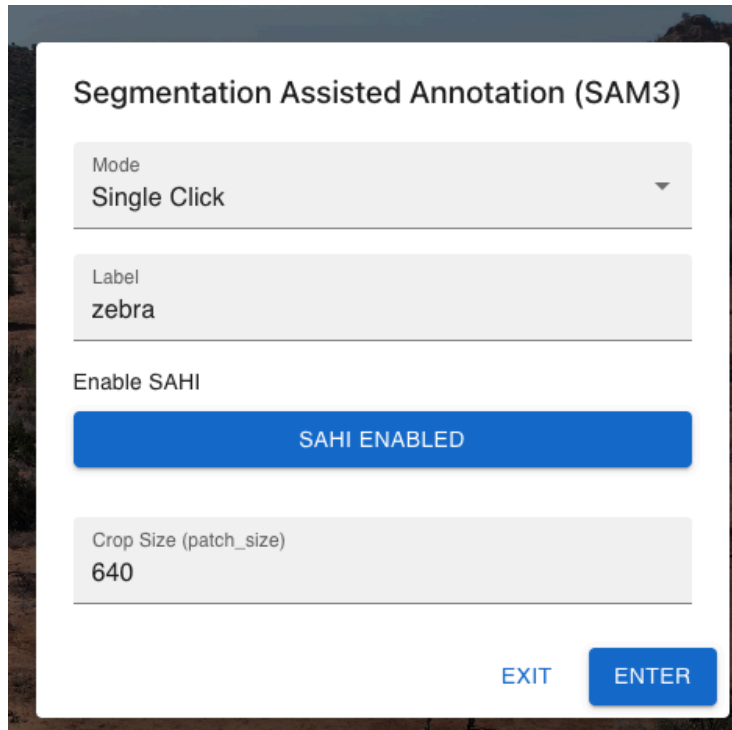
```
{
  "info": {
    "year": 2024,
    "version": "1.0",
    "date_created": "2024-06-01"
  },
  "licenses": [],
  "images": [
    {
      "id": 1,
      "width": 640,
      "height": 480,
      "file_name": "image1.jpg"
    }, ...
  ],
  "categories": [
    {
      "id": 1,
      "name": "animals",
      "supercategory": "none"
    },...
  ],
}
```

CANCEL

SAVE

SAM3 Segmentation-Assisted Annotation

SAM3 provides Single Click and Text Prompt modes to auto-generate bounding boxes. These modes support **SAHI (Slicing Aided Hyper Inference)** for improved detection of small objects in large images.



Segmentation Assisted Annotation (SAM3)

Mode
Single Click

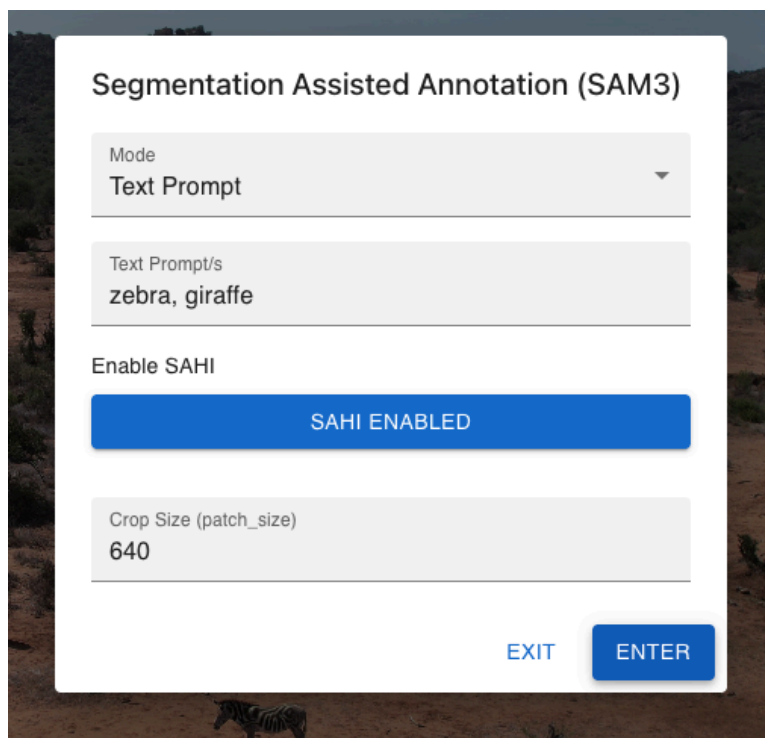
Label
zebra

Enable SAHI

SAHI ENABLED

Crop Size (patch_size)
640

EXIT ENTER



Segmentation Assisted Annotation (SAM3)

Mode
Text Prompt

Text Prompt/s
zebra, giraffe

Enable SAHI

SAHI ENABLED

Crop Size (patch_size)
640

EXIT ENTER

Step 2 — Generate Class Supports

Class supports are embedding representations of annotated object classes used as references during detection. Users select proposer and embedder models from the Patra catalog and specify patch sizes.

Annotation

Build Class Supports

Optimize Patch Size

Configure Detection Job

Visualize Proposals

Configure Classification Job

Object Classification

Generate Class Supports

Configure data paths, select a model card, and submit class-support generation.

Job Name

Generate Support embeddings for cow 2

Enter a custom name for your reference

System

cardinal-tapis

Job execution cluster

Source Image Directory

Browse

/fs/ess/PAS2699/farmers_meet/source

Enter full path to source image directory

Annotation File Path

Browse

/fs/ess/PAS2699/farmers_meet/annotation/cow.json

Enter full path to annotation file

Ground truth generated from previous step

Output Directory

Browse

/fs/ess/PAS2699/farmers_meet/output

Enter full path to output directory

Model Selection

Select Model

Patra Model cards - Select model(s)

DINOv3 ViT-S/16 (LVD-1689M Pretrained)

ViT-S/16 (DISTILLED FROM 7B TEACHER)

GATED

Small-sized DINOv3 Vision Transformer (ViT-S/16) distilled from the 7B teacher.

OWLv2 Base Patch16 Ensemble (zero-shot object detection)

1.0

Open-vocabulary object detector: text prompts → bounding boxes + scores. Used in our pipeline with GPU letterbox preprocessing, grounded post-processing, and eval JSON aligned to COCO-style boxes.

OWLv2 Large Patch14 Ensemble

ViT-L/14 ENSEMBLE (SELF-TRAINED)

GATED

Ensemble version of OWLv2 for state-of-the-art zero-shot object detection.

Selected Models (1)

OWLv2 Large Patch14 Ensemble

Crop/Patch Size

612

612 or [1024,612] (array of numbers ...) - Enter the list of patch sizes to try out to get a optimal proposal with highest IOU score

Device

CUDA

Method

Image

Click on generate class supports to start the job execution.

Generate Class Supports

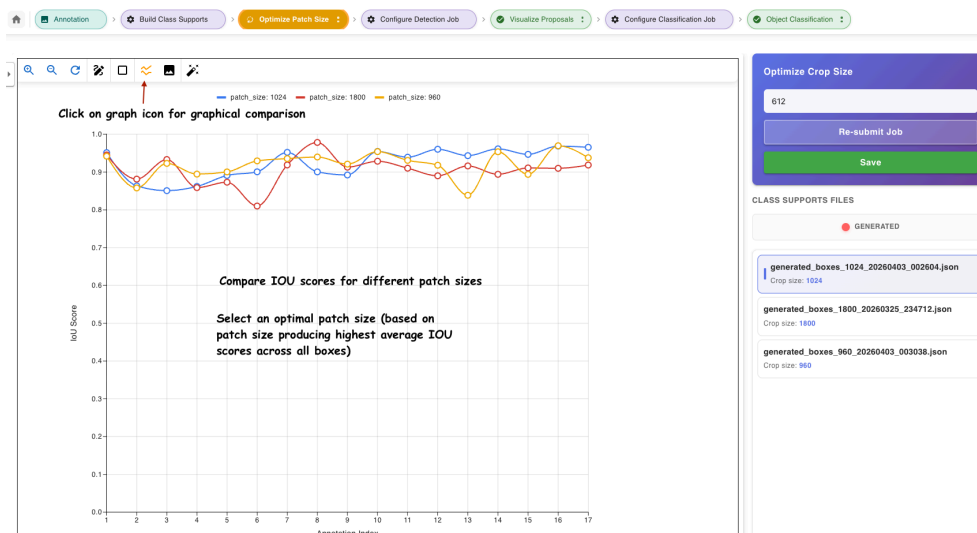
Step 3 — Optimize Patch Size

This step allows users to visually compare ground-truth annotations against model predictions for different patch sizes.

The interface shows a top navigation bar with tabs: Annotation, Build Class Supports, Optimize Patch Size (active), Configure Detection Job, Visualize Proposals, Configure Classification Job, and Object Classification. Below the navigation bar, a toolbar contains icons for search, zoom, pan, and other functions. The main area displays a grassy field with several cows, each enclosed in a red bounding box labeled 'cow'. A red arrow points from the 'Visualize the difference between ground truth and OWLv2 proposals' text to the field. On the right, a sidebar titled 'Optimize Crop Size' includes a text input field with '612', a 'Re-submit Job' button, and a 'Save' button. Below this, a section titled 'CLASS SUPPORTS FILES' shows a list of generated files: 'generated_boxes_1024_20260403_002604.json' (Crop size: 1024), 'generated_boxes_1800_20260325_234712.json' (Crop size: 1800), and 'generated_boxes_960_20260403_003038.json' (Crop size: 960). A red arrow points to the third file, and a blue text box says 'Click on different files to compare the proposals generated for different patch/crop size'.

IoU Graph

A multi-series line chart plots IoU scores against patch size, facilitating the identification of the optimal scale.



Step 4 — Configure Detection Job

Defines the query images and model parameters (e.g., SAHI settings, batch size, and thresholds) for the detection run.

Annotation

Build Class Supports

Optimize Patch Size

Configure Detection Job

Visualize Prompts

Configure Classification Job

Object Classification

Configure Detection Job

Detection Job Configuration

New Configuration

Job Name

Classification Job 1

Training System

ptcnn-taps

Query File Location

Browse

/Users/PAS2699/farmers_mee/query

Enter full path to query file or directory

Output Directory

Browse

/Users/PAS2699/farmers_mee/output3

Enter full path to output directory

Model Selection

Step 1: Select Proposer Model(s)

Select Model

OWLv2 Base Patch16 Ensemble (zero-shot object detection) 1.0

Open-vocabulary object detector: text prompts → bounding boxes + scores. Used in our pipeline with GPU letterbox preprocessing, grounded post-processing, and eval JSON aligned to COCO-style boxes.

OWLv2 Large Patch14 Ensemble 1.0

Open-vocabulary object detector: text prompts → bounding boxes + scores. Used in our pipeline with GPU letterbox preprocessing, grounded post-processing, and eval JSON aligned to COCO-style boxes.

Segment Anything Model 3 (SAM 3) 1.0

Foundation model for promptable segment representation (P2S).

Selected Models (2)

OWLv2 Large Patch14 Ensemble

Segment Anything Model 3 (SAM 3)

Clear All

Step 2: Select Embedder Model(s)

Select Model

BioCLIP 2 (via pybioclip) 2.1.1 (LIBRARY) / VIT-L/14 (MODEL) 1.0

Biology foundation model for taxonomic classification, trait prediction, and biodiversity analysis.

DINOv3 ViT-S/16 (LVD-1689M Pretrained) 1.0

Small-sized DINOv3 Vision Transformer (ViT-S/16) distilled from the 7B teacher.

OWLv2 Base Patch16 Ensemble (zero-shot object detection) 1.0

Open-vocabulary object detector: text prompts → bounding boxes + scores. Used in our pipeline with GPU letterbox preprocessing, grounded post-processing, and eval JSON aligned to COCO-style boxes.

Selected Models (3)

OWLv2 Large Patch14 Ensemble

BioCLIP 2 (via pybioclip)

DINOv3 ViT-S/16 (LVD-1689M Pretrained)

Clear All

Select existing configurations to view / edit

Saved Configurations

Click to load previous configurations

Classification Job 1

Classification Job 1

Models

Proposer: OWLv2 Large Patch14

Embedder: OWLv2 Large Patch14

Thresholds

Objectness: 0.59

Similarity: 0.20

NMS IoU: 0.50

Settings

Batch Size: 4

Device: CUDA

Method: Image

SAHI Tile: 960px, Overlap: 0.20

cow detection

cow detection

Models

Proposer: Segment Anything Model

Embedder: DINOv3 ViT-S/16 (LVD)

Thresholds

Objectness: 0.10

Similarity: 0.20

NMS IoU: 0.50

Settings

Batch Size: 4

Device: CPU

Method: Image

Advanced Settings

Batch Size

4

Device

CUDA

Method

Image

☒ Enable SAHI (Slicing Aided Hyper Inference)

SAHI Tile Size (pixels)

960

SAHI Overlap Ratio

0.2

Detection Thresholds

Objectness Threshold

0.59

Enter the confidence threshold (proposals below this threshold won't be considered as detections)

Similarity Threshold

0.2

If the similarity score is below this threshold, the detection is eliminated

NMS IoU Threshold

0.5

Threshold to eliminate overlapping boxes

Submit Detection Job

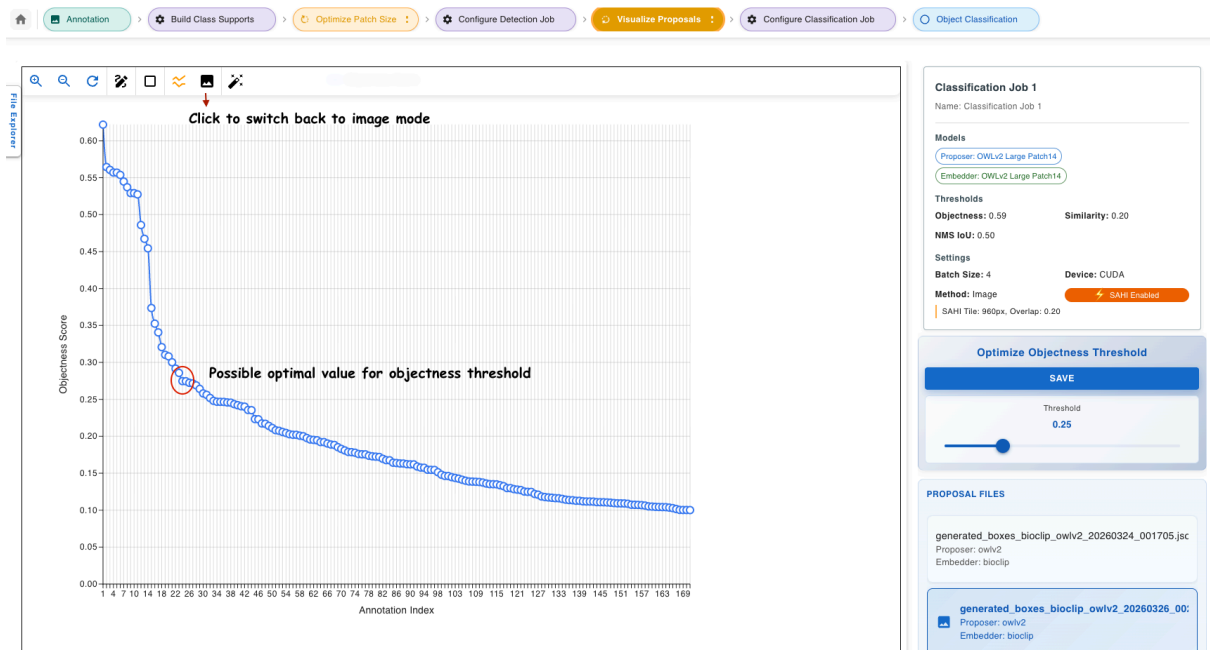
Step 5 — Visualize Proposals

Review raw object proposals before classification to find the optimal objectness threshold using real-time sliders.

The screenshot shows the 'Visualize Proposals' interface. The main area displays an aerial image with numerous red bounding boxes representing object proposals. Annotations include: 'Switch to graphical view' pointing to a button in the top toolbar, 'Save the threshold once you find the optimum value' pointing to a 'SAVE' button in the sidebar, and 'Select proposal file' pointing to a file selection button. The sidebar on the right contains configuration details for 'Classification Job 1', including models (Proposer: OWLv2 Large Patch14, Embedder: OWLv2 Large Patch14), thresholds (Objectness: 0.59, Similarity: 0.20, NMS IoU: 0.50), settings (Batch Size: 4, Device: CUDA, Method: Image, SAHI Tile: 960px, Overlap: 0.20), and a section for 'Optimize Objectness Threshold' with a slider set to 0.25 and a 'SAVE' button. Below this, 'PROPOSAL FILES' are listed, including 'generated_boxes_bioclip_owlv2_20260324_001705.jsc' and 'generated_boxes_bioclip_owlv2_20260326_00:'. A 'SAHI Enabled' status is also visible.

Objectness Score Graph

The graph view shows a descending line chart of objectness scores to help identify natural cutoffs.



Step 6 — Configure Classification Job

Maps proposal tensors to class support tensors and submits the classification job to the HPC cluster.

The screenshot shows the 'Configure Classification' web interface. At the top, there is a navigation bar with tabs: 'Optimize Patch Size', 'Configure Detection Job', 'Visualize Proposals', 'Configure Classification Job' (active), and 'Object Classification'. Below the navigation bar, the title 'Configure Classification' is displayed, followed by a subtitle: 'Configure classification parameters with embedder model, thresholds, and file paths.'

The main configuration area is divided into several sections:

- Basic Settings:** Includes a 'Job Name' field (containing 'Classification Job') and a 'System' dropdown menu (set to 'pitzer-lapis').
- Similarity Threshold:** A text input field with the value '0.2'. A note below it says 'Value between 0 and 1 for matching threshold'.
- Objectness Threshold:** A text input field with the value '0.59'. A note below it says 'Value between 0 and 1 for matching threshold'.
- Proposal and Class Support Mapping:** A section titled 'To map and compare proposal embeddings to support embeddings'. It includes a '+ Add Mapping' button and a list of mappings. The current mappings are:
 - features_owlv2_owlv2_20260326_002343.npz
 - class_supports_owlv2_1800_20260326_234712.npz
- Proposal Tensor File:** A text input field containing 'features_owlv2_owlv2_20260326_002343.npz'.
- Associated Class Support File:** A text input field containing 'class_supports_owlv2_1800_20260326_234712.npz'.

At the bottom right, there are 'Reset' and 'Submit Job' buttons.

Step 7 — Object Classification (Results)

View final labeled bounding boxes and export results as COCO JSON or Default JSON.

The screenshot shows the 'Object Classification (Results)' web interface. At the top, there is a navigation bar with tabs: 'Annotation', 'Build Class Supports', 'Optimize Patch Size', 'Configure Detection Job', 'Visualize Proposals', 'Configure Classification Job', and 'Object Classification' (active).

The main area is divided into two panels:

- Final detections with labels:** A large image showing a grassy field with several blue bounding boxes around objects. A red arrow points from the text 'Click here to download the final annotation file' to a 'DOWNLOAD' button.
- Classification Job 1:** A panel on the right showing job details. It includes:
 - Models:** 'Proposer: OWL v2 Large PatchNet' and 'Embedder: OWL v2 Large PatchNet'.
 - Thresholds:** 'Objectness: 0.59' and 'Similarity: 0.20'.
 - Settings:** 'APPS: 1000' and 'Batch Size: 4'.
 - Method:** 'Image'.
 - Device:** 'CUDA'.
 - Similarity Threshold:** A slider set to '0.85'.
 - DETECTION FILES:** A list of files for download:
 - 'detections_owlv2_0.2_0.2_20260327_153446.json' (Similarity Threshold: 0.2)
 - 'detections_owlv2_0.2_0.2_20260329_221145.json' (Similarity Threshold: 0.2)

Infrastructure & Integrations

- **Tapis HPC Jobs:** Compute-intensive tasks run on clusters like Pitzer or Expanse.
- **Tapis Vault:** Secure storage for Hugging Face tokens and credentials.
- **Patra Model Registry:** Live catalog access to ICICLE AI models.